NATHAN RIHET

Full Stack Engineer | Osaka, Japan

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PROFESSIONAL SUMMARY

Full Stack Engineer working with TypeScript, React, Next.js, Python and FastAPI. My recent work includes computer-vision applications, GPU job scheduling and background workflows, with earlier experience in enterprise generative AI and freelance web development.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Frontend: React, Next.js

Backend & Data: FastAPI, Node.js, PostgreSQL, MongoDB, REST APIs

Cloud & Delivery: AWS, Azure, Docker, Terraform, GitHub Actions, CI/CD

Workflows & Testing: Temporal, pytest, Cypress, integration and end-to-end testing

AI Applications: Computer vision integration, GPU inference workflows, RAG

Spoken Languages: French (native), English (professional), Japanese (elementary)

PROFESSIONAL EXPERIENCE

ROKKEN

Full Stack Engineer | Osaka, Japan | Sep 2025 - Present

- Built the full-stack application around existing video-analysis pipelines: resumable uploads, secure playback, versioned human review and access-controlled sharing.
- Designed and implemented a PostgreSQL-backed GPU scheduler with fair capacity sharing between tenants, bounded admission, idempotent requests and worker draining.
- Build background workflows with Temporal and PostgreSQL, covering retries, cancellation and failure recovery.
- Expand pytest and Cypress coverage and improve GitHub Actions pipelines with targeted test execution and Docker build caching.

FREELANCE

Full Stack Developer | Remote | May 2024 - Aug 2025

- Delivered client web applications, covering requirements, custom APIs, authentication, dashboards and deployment.
- Built and launched DocsRetriever, a B2B document-search SaaS using Next.js, NestJS, FastAPI and MongoDB. Maintained and supported it through Aug 2025; now offline.

CAPGEMINI

Software Engineer | Valbonne, France | Apr 2024 - Mar 2025

- Developed enterprise generative-AI applications with React, Python, FastAPI and LangChain, first during an internship and then in a full-time role.
- Migrated backend services from Flask to FastAPI, reducing p95 latency from 500 ms to 200 ms.
- Implemented hybrid document retrieval and automated Azure deployments with Docker, Terraform and CI/CD.

CAWITA TECHNOLOGIES

Full Stack Developer | Antibes, France | Sep 2022 - Aug 2023

- Developed and maintained full-stack applications and REST APIs with MongoDB, Express, Angular and Node.js.
- Automated deployment workflows with Jenkins, Bash and Linux-based infrastructure.

TEACHING EXPERIENCE

UNIVERSITÉ CÔTE D'AZUR
Academic Instructor - Conversational Agents | Nice, France | Dec 2024 - Jan 2025

- Designed and delivered seven practical workshops to more than 20 Master's students.
- Created hands-on material covering Python, FastAPI, LangChain, REST APIs and LLM application development.

Course: github.com/NathanKneT/Master-AI-Chatbot-Course-2024

EDUCATION

Master's in Applied Artificial Intelligence (MIAGE) | Université Côte d'Azur | 2022 - 2024

Computer Science Exchange - AI Specialization | Université Laval | 2023 - 2024

Bachelor's in Computer Methods Applied to Business Management | Université Côte d'Azur | 2021 - 2022

SELECTED ENGINEERING PROJECTS

Kororo | Browser-based 3D puzzle game | Playable prototype

Stack: TypeScript, React, Next.js, Three.js, React Three Fiber, Vitest

- Built a 3D puzzle game where players push and pull blocks to create a path, with a deterministic simulation separated from rendering.
- Implemented undo and replay, level solving and validation in Web Workers, Vitest tests, keyboard/touch controls and adaptive rendering quality.

Demo: nathanknet.itch.io/kororo-private-playtest

RTHT-3D | Python, MediaPipe, OpenCV, Blender, UDP | github.com/NathanKneT/RTHT-3D

- Built a real-time dual-hand tracking interface connecting Python and Blender through UDP.
- Produced a technical demonstration that reached 343,000 Instagram views, recorded in August 2026.

Demo: instagram.com/reel/DJLriUMSSpy

LEADERSHIP & CREATIVE PRACTICE

MIAGE Student Association - President: Led student initiatives and coordinated technical workshops and events.

Photography & Portfolio: Maintain a Gatsby 5 portfolio covering portrait, urban, night and live-performance photography. nathanglhf.com